

Aberrant hippocampal Ca^{2+} micro-waves following synapsin-dependent adeno-associated viral expression of Ca^{2+} indicators

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Abstract

Genetically encoded calcium indicators (GECIs) such as GCaMP are invaluable tools in neuroscience to monitor neuronal activity using optical imaging. The viral transduction of GECIs is commonly used to target expression to specific brain regions, can be conveniently used with any mouse strain of interest without the need for prior crossing with a GECI mouse line and avoids potential hazards due to the chronic expression of GECIs during development. A key requirement for monitoring neuronal activity with an indicator is that the indicator itself minimally affects activity. Here, using common adeno-associated viral (AAV) transduction procedures, we describe spatially confined aberrant Ca^{2+} micro-waves slowly travelling through the hippocampus following expression of GCaMP6, GCaMP7 or R-CaMP1.07 driven by the synapsin promoter with AAV-dependent gene transfer, in a titre-dependent fashion. Ca^{2+} micro-waves developed in hippocampal CA1 and CA3, but not dentate gyrus (DG) nor neocortex, were typically first observed at 4 weeks after viral transduction, and persisted up to at least 8 weeks. The phenomenon was robust, observed across laboratories with various experimenters and setups. Our results indicate that aberrant hippocampal Ca^{2+} micro-waves depend on the promoter and viral titre of the GECI, density of expression as well as the targeted brain region. We used an alternative viral transduction method of GCaMP which avoids this artifact. The results show that commonly used Ca^{2+} -indicator AAV transduction procedures can produce artefactual Ca^{2+} responses. Our aim is to raise awareness in the field of these artefactual transduction-induced Ca^{2+} micro-waves and we provide a potential solution.

Impact statement: Common AAV transduction procedures induce artefactual spatially confined Ca^{2+} waves in the hippocampus.

eLife assessment

This **important** study provides **convincing** evidence of artifactual calcium micro-waves during calcium imaging of populations of neurons in the hippocampus using methods that are common in the field. The work raises awareness of these artifacts so that any research labs planning to do calcium imaging in the hippocampus can avoid them by using alternative strategies that the authors propose.

Introduction

There has been an explosion in the use of imaging techniques to record neuronal activity over the past 30 years, starting with the introduction of organic calcium indicators to measure neuronal population activity (Yuste & Katz 1991 [↗](#)) and accelerated by rapid advances in the development of genetically encoded Ca^{2+} indicators (GECIs) (Miyawaki *et al.* 1997 [↗](#)). Specific advantages of neuronal Ca^{2+} imaging with GECIs lie in the ability of chronic cellular scale recordings of sizeable, densely labelled neuronal or glial populations with subtype specificity, without having to perturb the cell membrane or add synthetic chemical to the brain (Grienberger & Konnerth 2012 [↗](#); Rose *et al.* 2014 [↗](#); Semyanov *et al.* 2020 [↗](#)).

Commonly used GECIs such as the GCaMP family have been continually improved since their initial development (Nakai *et al.* 2001 [↗](#)), offering high signal to noise ratio, sensitivity and response kinetics such that they can detect single action potentials in vivo. This allows the reporting of cellular activity as well as the activity of sub-compartments such as the dendritic arbour (Chen *et al.* 2013 [↗](#); Dana *et al.* 2019 [↗](#); Zhang *et al.* 2023 [↗](#)). Typically, GCaMP is expressed using transgenic animals or adeno-associated viral (AAV) transduction techniques (Tian *et al.* 2012 [↗](#) also see Grødem *et al.* 2023 [↗](#)). The use of transgenic animals has the advantage of not requiring AAV transduction, thus reducing surgery load for animals and likelihood of indicator overexpression. In contrast, AAV GECI transduction is straightforward (breeding/crossing not required), can be targeted to virtually any brain region, and typically offer enhanced fluorescence (due to higher expression levels). Further, AAV transduction avoids potential hazards due to chronic GECI expression during development.

While offering unprecedented new insights into cellular scale neuronal network dynamics, it has also been reported that GECI expression in neurons can result in unwanted side-effects. Depending on the expression approach, neurons have shown reduced dendritic branching and impairment in cell health leading to cytotoxicity and cell death (Gasterstädt *et al.* 2020 [↗](#); Resendez *et al.* 2016 [↗](#)). Further, increased Ca^{2+} buffering due to addition of Ca^{2+} indicators has been associated with alterations in intracellular Ca^{2+} dynamics (Grienberger & Konnerth 2012 [↗](#); McMahon & Jackson 2018 [↗](#)). In addition, chronic expression of GCaMP can lead to accumulation in the nucleus and changes in gene expression (Yang *et al.* 2018 [↗](#)). Again, depending on the specific expression approach, GCaMP variant and experimental time course, such changes may alter cellular physiology and excitability. For example, increased firing rates have been observed in hippocampal neurons expressing GCaMP5G from CaMKIIa-Cre; PC::G5-tdT mice, and epileptiform activity in neocortex in some GCaMP6 expressing transgenic mice (Gee *et al.* 2014 [↗](#); Steinmetz *et al.* 2017 [↗](#)).

Here, we describe micro-scale Ca^{2+} waves that are highly confined in space, and progress slowly through the hippocampus following local GCaMP or R-CaMP viral transduction. Such aberrant hippocampal waves were typically first observed 4 weeks following injection of commercially available AAVs expressing GCaMP6, GCaMP7 or R-CaMP1.07 under the synapsin promoter. The

phenomenon occurred upon GECI transduction in CA1 and CA3, but not in dentate gyrus (DG) nor neocortex, was robustly observed across laboratories with various experimenters and set-ups, and highlights the necessity of careful use of transduction methods and control measures. Reducing the transduction titre diminished the likelihood of aberrant hippocampal Ca^{2+} waves, and an alternative viral transduction method employing sparser and Cre-dependent GCaMP6s expression in principal cells avoided the aberrant Ca^{2+} waves. Further, in three transgenic GCaMP mouse lines (thy1-GCaMP6s or 6f; Vglut1-IRES2-Cre-D x Ai162(TIT2L-GC6s-ICL-tTA2), aberrant Ca^{2+} micro-waves were never observed. The aim of this article is to raise awareness in the field of artefactual transduction-induced Ca^{2+} waves and encourage others to carefully evaluate their Ca^{2+} indicator expression approach before embarking on chronic in vivo calcium imaging of the hippocampus.

Results

Aberrant Ca^{2+} micro-wave progression through hippocampus

Based on published protocols, we injected AAV1 particles (pAAV.Syn.GCaMP6s.WPRE.SV40, Addgene #100843, titre 1×10^{13} vg/ml) in the hippocampus (total injection vol.: 500nl undiluted [1:1 virus solution) of C57BL6 wild-type animals (6 weeks old), and performed in vivo two-photon imaging to record cellular activity at 2, 4 and 6-8 weeks (wks) post-injection (p.i.) (**Fig. 1a**). Viral transduction resulted in GCaMP6s expression throughout the hippocampal CA1, CA3 and DG areas under the imaging window (**Fig. 1b**). As expected, the expression was primarily restricted to the ipsilateral hippocampus, with some labelling of projection pathways also in the contralateral hippocampus. There was no evidence of gross transduction-related morphological changes to the hippocampus, (see **Fig. 1b**) with no changes in CA1 pyramidal cell layer thickness or CA1 thickness (pyramidal layer thickness: 49 ± 12.5 μm ipsilateral and 50.3 ± 11.1 μm contralateral, $n=4$, Student's t-test $p=0.89$; CA1 thickness: 553.3 ± 14 μm ipsilateral and 555.8 ± 62 μm contralateral, $n=4$, Student's t-test $p=0.94$; 48 ± 13 weeks post injection at time of perfusion). At 4 wks after injection, a time point commonly used for imaging cellular activity, we observed distinctive aberrant microscale Ca^{2+} waves that travelled through CA1 recruiting neighbouring cells (**Fig. 1c-d**, **Supplementary video 1**, $n=4$ mice). Ca^{2+} micro-waves were maintained up to 6-8 wks after AAV injections (**Fig. 1e**, $n=4$ mice). In wildtype mice, these Ca^{2+} micro-waves were not observed at an earlier timepoint (2 wks p.i.; $p<0.05$ using Kruskal-Wallis H-test for comparison between the 3 time points).

The properties of the Ca^{2+} micro-waves depended on the hippocampal region and exact recording location. For instance, although the Ca^{2+} waves were consistently observed in CA1, the spatial dimensions of the Ca^{2+} micro-waves were broader in the stratum oriens compared with stratum pyramidale of CA1 (**Supplementary videos 4 [str. pyr.] and 5 [str. oriens]**), which likely reflects concomitant neuropil activation. We next examined, if the CA1 network is particularly prone to the generation of such waves and if they show regional specificity. Upon viral GCaMP6s transduction under synapsin, Ca^{2+} waves were observed in both CA1 ($n=4/4$; **Supplementary videos 1-5**) and CA3 ($n=1/1$; **Supplementary video 6**), but interestingly, not in the dentate gyrus ($n=3$ mice, 4, 8 & 10 wks p.i., 40 min total recording time per mouse). In contrast to hippocampus, synapsin dependent GCaMP6s expression restricted to the neocortex (V1 or somatosensory cortices) did not result in cortical Ca^{2+} waves in our hands (**Fig. 1f**, $n = >20$ mice).

Aberrant Ca^{2+} microwaves in disease models

The observed Ca^{2+} micro-waves were distinct from local seizure activity (no rhythmicity, no typical ictal evolution, no postictal depression) (Masala *et al.* 2022; Muldoon *et al.* 2015; Wenzel *et al.* 2017, 2019a) and spreading depolarization/depression phenomena (no concentric expansion, no post-wave neural depression). However, the occurrence of these artificial events may be confused as aberrant activity related to a pathology, especially when

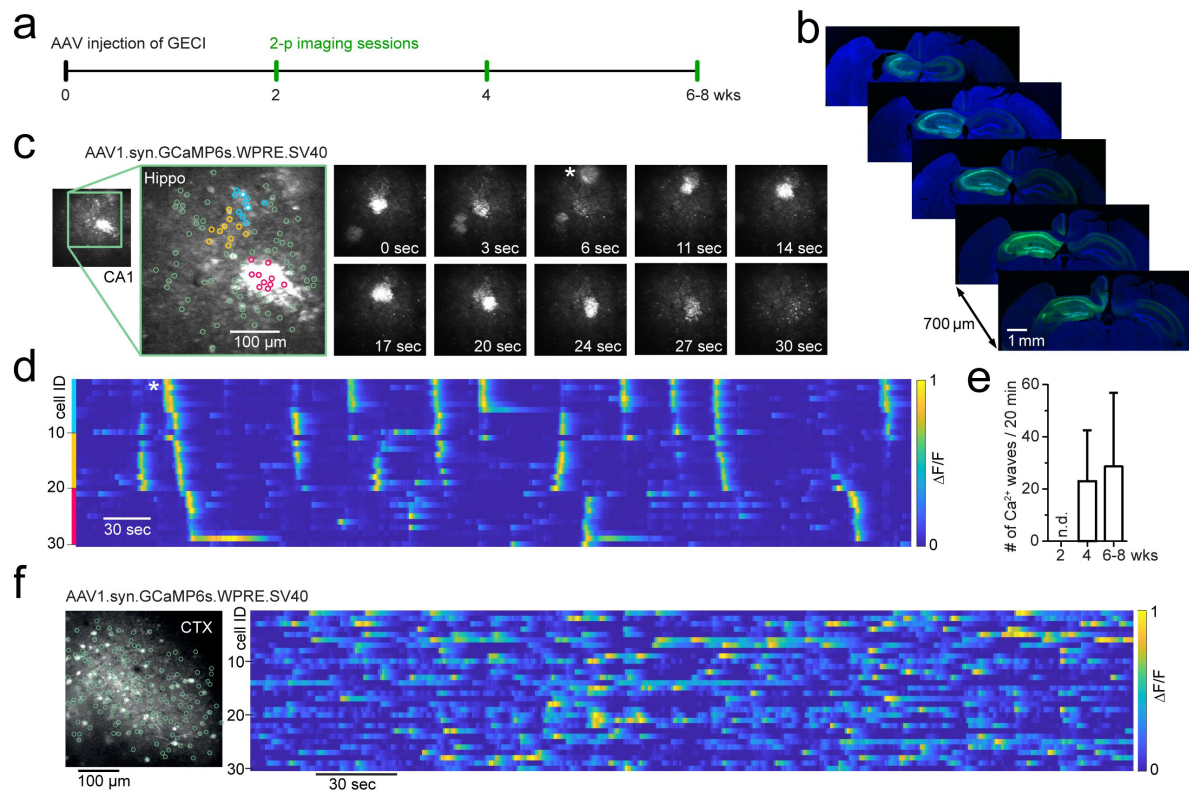


Figure 1

Development of Ca^{2+} micro-waves travelling through hippocampus following GCaMP transduction.

a: Experimental protocol to examine CA1 neuronal activity using two-photon imaging following AAV transduction of genetically encoded Ca^{2+} indicators. **b:** Immunohistochemical sections following last imaging session. GCaMP6s (AAV1.syn.GCaMP6s.SV40, addgene #100843) expression throughout the ipsilateral hippocampus and projection pathways in the contralateral hippocampus. **c:** Two-photon Ca^{2+} imaging of FOV in CA1 at 4wks p.i. showing aberrant Ca^{2+} micro-waves (see also Supplementary video 1). Magnified inset shows 3 colored neuronal subgroups (blue, orange, magenta) based on their spatial vicinity from a total population of 100 identified neurons (green). *right:* time series of two-photon Ca^{2+} imaging FOVs showing two Ca^{2+} micro-waves, the first at 0 sec, the second appearing at 6 sec (asterisk). The second wave progresses through FOV over dozens of seconds. **d:** Raster-plot of individual neuronal Ca^{2+} activity ($\Delta F/F$, 1min moving window, traces max-normalized per neuron) from neighboring subgroups (colors correspond to c). Asterisk (same as in c): a Ca^{2+} micro-wave advances through neighboring neuronal subgroups. **e:** Occurrence rate of aberrant Ca^{2+} micro-waves with increasing expression time, following viral transduction of AAV1.syn.GCaMP6s.SV40 in mature C57BL6 wild-type animals. n.d. = none detected. **f:** Two-photon Ca^{2+} imaging FOV in the visual cortex at 6wks p.i. (left) with normal sparse spontaneous Ca^{2+} activity and no detected Ca^{2+} micro-waves (right; raster plot of $\Delta F/F$, 1min moving window, traces max-normalized per neuron).

studying pathologies with known cellular and network hyperexcitability. For example, we initially found the aberrant hippocampal Ca^{2+} micro-waves in the *Scn2a*^{A263V} model of genetic epilepsy; however, these Ca^{2+} waves in CA1 of heterozygous animals (5/5 mice) were in general similar to those detected in wildtype animals at 4 wks p.i.. In the *Scn2a*^{A263V} model, in one case (1/5 animals), Ca^{2+} waves were observed even at 2 wks p.i. (**Supplementary video 2**). Furthermore, hippocampal transduction of jGCaMP7f under synapsin (Addgene #104488, AAV9 particles, original titre 2.5×10^{13} vg/ml, total inj. vol. 1000nl [1:2 dilution]) in a mouse model of Alzheimer's disease (PV-Cre::APPswe/PS1dE9) also resulted in Ca^{2+} micro-waves (n=3/6 mice). Together, these experiments show that common AAV injection procedures of GECIs under the synapsin promoter lead to artefactual hippocampal Ca^{2+} micro-waves in wildtype mice and genetic mouse models of disease.

Properties & robustness of aberrant hippocampal Ca^{2+} waves

Next, we investigated the robustness of the aberrant Ca^{2+} micro-waves across institutes and conditions. We chose to compare the incidence of aberrant Ca^{2+} micro-waves in the CA1 region in four separate institutes in three different countries following transduction of GCaMP6s (Addgene #100843; IEECR/UoB, CU), GCaMP6m or jGCaMP7f (Addgene #100841 or #104488; DZNE) or RCaMP1.07 (Viral Vector Facility UZH #V224-9; UZH, **Supplementary video 3**)(**Table 1** [↗](#)).

The incidence of aberrant hippocampal Ca^{2+} micro-waves was robust, observed at the 4 different institutes each using variations of commonly used, published viral transduction procedures and standard two-photon Ca^{2+} imaging protocols (**Table 1** [↗](#) and see methods for more details). Importantly, aside from the targeted region, the viral titre was important, as halving the original AAV1.syn.GCaMP6m viral titre decreased the number of animals that developed aberrant Ca^{2+} micro-waves from 80% of animals (4/5, original titre, 1×10^{13} vg/mL) to 43% of animals (3/7, 50% reduced titre, 0.5×10^{13} vg/ml) (see **Table 1** [↗](#)). To statistically test the involvement of expression level, we used a generalized linear model. For injections into CA1 in the hippocampus (n=28), a multivariate logistic GLM (Ca^{2+} wave ~ dilution + p.i. wks) found both dilution and post injection weeks were significantly related to Ca^{2+} wave incidence (model deviation above null = 7.5; dilution: z score = 2.18, p < 0.05; p.i. wks: z score = 2.22, p < 0.05).

To examine how robust the Ca^{2+} micro-waves properties were, we compared properties across the different laboratories following expression of GCaMP6m and GCaMP6s variants. The occurrence rate of the aberrant Ca^{2+} micro-waves was similar across the different institutes (**Fig. 2a** [↗](#)). Ca^{2+} micro-waves were spatially confined (diameter range of 200-300 μm), moved across the field of view with slow progression speeds (speed range of 5-25 $\mu\text{m}/\text{sec}$) (**Fig. 2b** [↗](#)), and displayed no rhythmicity but rather plateau-like Ca^{2+} activity. No statistical differences were observed in Ca^{2+} micro-wave properties between the different institutes suggesting that these values provide a reasonable range. In addition, the Ca^{2+} micro-waves were not restricted to a single GECI variant or version, with Ca^{2+} waves observed following expression of GCaMP6m (n=4), GCaMP6s (n=5), and GCaMP7f (n=3), as well as R-CaMP1.07 (n=1)(**Fig. 2c** [↗](#) and **Table 1** [↗](#)).

In summary, upon synapsin-promoter-dependent AAV Ca^{2+} indicator expression, depending on time of expression and viral transduction titre, Ca^{2+} micro-waves were specifically observed in the CA1 and CA3 subregions of the hippocampus. For CA1, the Ca^{2+} micro-waves were observed across laboratories and countries and animal models, using common transduction procedures (for overview see **Table 1** [↗](#)).

AAV	Construct	Source (Addgene id)	Orig. titre [vg/ml]	Dilution	Inj. vol.	p.i Wks	Region	Ca ²⁺ -wave incidence (%)	n	Mouse model	Institute
AAV1	Syn.GCaMP6s	100843	1x10 ¹³	1:1	0.5	2	CA1	0	0/4	wt	UoB
AAV1	Syn.GCaMP6s	100843	1x10 ¹³	1:1	0.5	4-6	CA1	100	4/4	wt	UoB
AAV1	Syn.GCaMP6s	100843	1 x10 ¹³	1:1	0.5	2	CA1	20	1/5	Scn2a ⁺	UoB
AAV1	Syn.GCaMP6s	100843	1 x10 ¹³	1:1	0.5	4-8	CA1	100	5/5	Scn2a ⁺	UoB
AAV1	Syn.GCaMP6m	100841	1 x10 ¹³	1:1	1	8	CA1	80	4/5	wt	DZNE
AAV1	Syn.GCaMP6m	100841	1 x10 ¹³	1:2	1	6	CA1	43	3/7	wt	DZNE
AAV1	Syn.GCaMP6s	100843	1 x10 ¹³	1:2	0.5	4-10	DG	0	0/3	wt	UoB
AAV9	Syn.jGCaMP7f	104488	2.5 x10 ¹³	1:2	1	10-14	CA1	50	3/6	APPswe [#]	DZNE
AAV1	Syn.GCaMP6s	100843	1 x10 ¹³	1:1	0.25	3-12	CA3	100 (1 exp.)	1/1	wt	CU
AAV1	Syn.GCaMP6s	100843	1 x10 ¹³	1:1	0.25	3-5	CA1	100 (1 exp.)	1/1	wt	CU
AAV1	Syn.GCaMP6s	100843	1 x10 ¹³	1:2	0.8	4-5	Ctx	0	0/>20	wt	CU
AAV1	Syn.GCaMP6f	100837	7x10 ¹²	1:2	0.75	3-6	Ctx	0	0/>20	wt	CU
AAV9	hSyn1.R-CaMP1.07	V224-9 ^{\$}	4.3x10 ¹²	1:1	0.2	8	CA1	100	2/2	wt	UZH
AAV1	syn.Flex.GCaMPm	100838	1 x10 ¹³	1:2	0.5	34-38	CA1	0	0/3	wt	DZNE
AAV9	CamKII0.4.Cre.SV40	105558	1x10 ¹³	1:2	0.5	6	CA1	0	0/2	wt	UoB
AAV1	syn.Flex.GCaMP6s	100845	1 x10 ¹³	1:2	0.5	6	CA1	0	0/2	wt	UoB
AAV1	CamKII0.4.Cre.SV40	105558	1x10 ¹³	1:2	0.5	6	CA1	0	0/2	wt	UoB

Table 1

Viruses used for expression of GEC1.

Viral titre is from Addgene documentation and was used at original concentration (dilution of 1:1) or at a dilution of 1:2. Syn.Flex.GCaMP6s and CamKII0.4.Cre were co-injected and therefore diluted to 1:2. Two-photon Ca²⁺ imaging was performed from 2 weeks after injection in the hippocampus (CA1, CA3 or DG) or neocortex (Ctx). Ca²⁺ micro-wave incidence was determined from the number of animals exhibiting Ca²⁺ micro-waves at the specified timepoint and region. * in heterozygous *Scn2a*^{A263V} mice. # in PV-Cre::APPswe/PS1dE9 mice. \$ Sourced from Viral Vector Facility University of Zurich (VVF/UZH). Inj. vol. = injection volume in µl; p.i. = post-injection; wks = weeks.

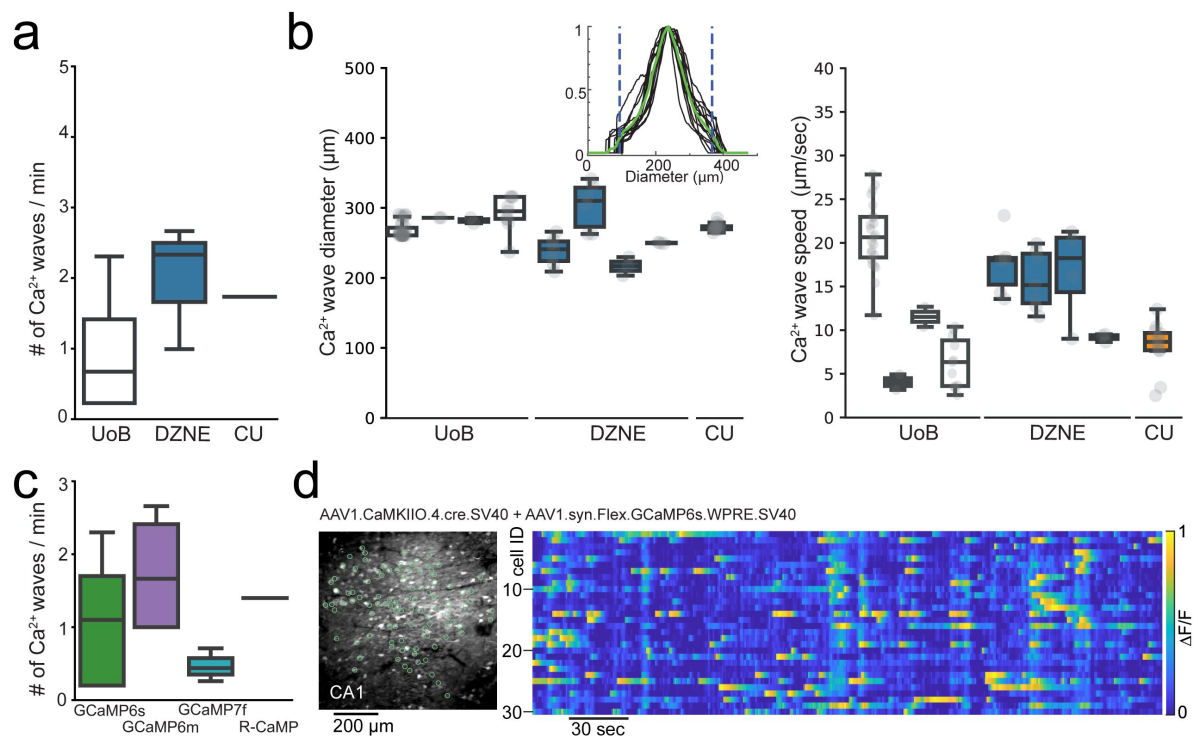


Fig. 2

Aberrant Ca^{2+} micro-waves are consistent across laboratories and GECI variant.

a: Plot of the occurrence rate of aberrant Ca^{2+} micro-waves in CA1 at the different institutes at 6-8wks after injection of GCaMP6s or GCaMP6m. **b:** Ca^{2+} micro-wave diameters (left) and progression speed (right) in CA1 from each animal recorded across institutes. Inset: Histogram of fluorescent intensity taken across each Ca^{2+} -wave within an animal. Green line is the average, areas outside dashed lines mark 10% lowest fluorescence values, which were excluded from analysis. **c:** Plot of the occurrence rate of aberrant Ca^{2+} micro-waves in CA1 following injection with commonly used GECIs (see [table 1](#)). **d:** Two-photon Ca^{2+} imaging FOV (left) in hippocampal CA1 following dual injection approach for conditional GCaMP6s expression (6 wks p.i.) with normal sparse spontaneous Ca^{2+} activity and no detection of Ca^{2+} micro-waves (right; raster plot of $\Delta F/F$, 1min moving window, traces max-normalized per neuron).

Alternative transduction method of

GCaMP to avoid aberrant Ca^{2+} micro-waves

In light of these results, we tested an alternative expression approach to avoid aberrant hippocampal Ca^{2+} micro-waves. To this end, we selected an approach to both limit the expression to principal cells and label a sparse population of the principal cells using a dual AAV injection approach. Here, Cre-dependent expression of GCaMP6s or GCaMP6m was achieved in a sparse population of principal cells under the CaMKII promoter (AAV1.syn.Flex.GCaMP6s.WPRE.SV40, Addgene #100845, and AAV1.CamKII0.4.Cre.SV40, Addgene #105558; $n=2$ or, AAV1.Syn.Flex.GCaMP6m.WPRE.SV40, Addgene #100838 and AAV9.CamKII0.4.Cre.SV40, Addgene #105558; $n=3$; **Fig. 2d**) (see Hare *et al.* 2022; Jimenez *et al.* 2020; Sheffield & Dombeck 2015), upon which no Ca^{2+} micro-waves were observed (0/5 animals, **Fig. 2d**). Further, hippocampal Ca^{2+} micro-waves were neither observed in transgenic thy1-GCaMP6s nor 6f mice (JAX strain 025776 or 024276; up to 3 months of chronic imaging in $n>30$ mature mice age $>p60$, cumulative imaging time $>200\text{hrs}$), nor in Vglut1-IRES2-Cre-D mice crossed with Ai162(TIT2L-GC6s-ICL-tTA2)-D mice (JAX strains: 037512, 031562; up to 3 months of chronic imaging in $n=5$ mature mice $>p60$).

Discussion

Here we report titre- and expression-time-dependent aberrant hippocampal Ca^{2+} micro-waves in CA1 and CA3 regions following viral expression of GCaMP or R-CaMP1.07 under the synapsin promoter. These aberrant Ca^{2+} micro-waves robustly occurred, were observed in 4 different institutes each using a common viral transduction approach and standard two-photon Ca^{2+} imaging protocols.

Ca^{2+} micro-waves were typically first detected at $\sim 4\text{wks}$, rarely also at 2 wks, after injection. Thus, there may be a time window when Ca^{2+} activity could be recorded in the absence of this artefactual phenomenon. However, we would still hesitate to use this specific approach for hippocampal imaging experiments as, although unknown from our data, more subtle alterations may occur prior to visible onset of aberrant activity. Further, at sites more distal to the injection site with lower expression levels, Ca^{2+} micro-waves may not be observed; however, it may very well be that Ca^{2+} micro-waves in regions with higher expression will affect fine-scaled neuronal population dynamics in primarily unaffected neighbouring regions.

The presence of Ca^{2+} micro-waves was not restricted to a single GCaMP variant or version, and was observed using either GCaMP6m, GCaMP6s, or GCaMP7f. The phenomenon was also observed upon transduction of R-CaMP1.07, indicating that these aberrant hippocampal waves are not restricted to GCaMP indicators, but rather present a general phenomenon following Ca^{2+} -indicator transduction. Notably, the viral transduction titre was a key factor, as reducing the viral transduction titre from 1×10^{13} vc/ml (500nl or 1000nl of a 1:1 undiluted virus solution) to 5×10^{12} vc/ml (1000nl 1:2 solution, single injection) decreased, albeit did not yet prevent, the occurrence of Ca^{2+} micro-waves. In the literature, hippocampal GCaMP transduction procedures in mice typically include one to several separate nearby injections, with a total volume of transduced undiluted virus ranging from 60nl to 500nl (Cai *et al.* 2016; Jimenez *et al.* 2020; Keinath *et al.* 2022; Pettit *et al.* 2022; Radvansky *et al.* 2021; Skoczek *et al.* 2018; Szabo *et al.* 2022; Weisenburger *et al.* 2019; Wirtshafter & Disterhoft 2022; Zaremba *et al.* 2017). In other studies, syn-GCaMP virus was diluted prior to injection (up to 1:10) (Jimenez *et al.* 2020; Zong *et al.* 2022), resulting in varied transduction volumes up to 1500nl. In our hands, a reduced viral titre of 5×10^{12} vc/ml in a 1000nl injection volume still resulted in aberrant Ca^{2+} micro-waves. Thus,

viral transduction titres per volume well below this number and diluted transduction solutions are advisable for syn-GCaMP expression in the hippocampus, if AAV syn-Ca²⁺-indicator transduction is desired for a planned in vivo hippocampal imaging experiment.

If possible, alternate viral GCaMP expression approaches should be chosen. As a possible alternative, similar to previous reports using dual AAV injections or AAV in Cre-driver mouse lines (Farrell *et al.* 2020 [↗](#); Grosmark *et al.* 2021 [↗](#); Mineur *et al.* 2022 [↗](#); Rolotti *et al.* 2022 [↗](#); Terada *et al.* 2022 [↗](#)), we find that Cre-dependent AAV GCaMP expression (IEECR/UoB) in pyramidal neurons does not cause aberrant hippocampal Ca²⁺ micro-waves. Moreover, we have not observed this aberrant phenomenon in transgenic thy1-GCaMP6s or 6f mice (JAX strain 025776 or 024276) (Masala *et al.* 2022 [↗](#); Rupprecht *et al.* 2023 [↗](#); Wenzel *et al.* 2019b [↗](#)), nor in Vglut1-IRES2-Cre-D x Ai162(TIT2L-GC6s-ICL-tTA2)-D mice (JAX strains: 037512, 031562). It goes beyond the scope and available resources in our laboratories to further identify, which viral GCaMP transduction approaches avoid the reported phenomenon. It seems likely that the underlying mechanisms for this artefact comprise transduction titre and time period of GCaMP or R-CaMP1.07 expression, region specificity and density of expression. Importantly, although our data suggest some regions and AAV constructs seem more prone to generate artefactual Ca²⁺ waves under certain conditions, this does not mean that Ca²⁺ waves cannot be generated in other regions, or with other constructs or promoters. It remains unclear if the observed phenomenon is restricted to Ca²⁺ indicator viral expression in mice, or if it extends to different animal models as well. In this regard, a previous report did not observe Ca²⁺-waves in rats following synapsin-dependent GCaMP6m expression, although notably, imaging was performed under isoflurane anaesthesia (Sosulina *et al.* 2021 [↗](#)). Further, disentangling the exact cellular mechanisms of the phenomenon from technical aspects is difficult as e.g. the mere change in the transduction procedure will affect GECI expression level. For instance, although Ca²⁺ waves were not observed following conditional expression of GCaMP with CaMKII.Cre, which may suggest a requirement for interneuronal expression, it may also simply reflect differences in final GCaMP expression density and levels between the two transduction procedures.

In the context of this study, the phenomenon of Ca²⁺ micro-waves is possibly related to the expression of exogenous Ca²⁺ buffer and the resulting effects on Ca²⁺ dynamics and gene expression (McMahon & Jackson 2018 [↗](#); Rose *et al.* 2014 [↗](#); Yang *et al.* 2018 [↗](#)), which may be why our findings extended across genetically encoded Ca²⁺ indicators. Beyond clearly being abnormal, the exact nature of the observed Ca²⁺ micro-waves remains unclear and may reflect Ca²⁺ influx during action potential firing or possibly Ca²⁺ release from internal stores. In a limited dataset, we tried to detect the Ca²⁺ micro-waves by hippocampal LFP recordings (insulated Tungsten wire, diameter ~110µm). We could not identify a specific signature, e.g. ictal activity or LFP depression, which may correspond to these Ca²⁺ micro-waves. The shortcoming of these LFP recordings is that we could not simultaneously perform hippocampal 2-photon microscopy and thus, it is uncertain if the Ca²⁺ micro-waves indeed occurred in proximity to our electrode. We did not evaluate the effect of Ca²⁺ micro-waves on physiological activity. Based on the data presented here, it appears reasonable to hypothesize that such waves obscure if not interfere with physiological activity e.g. with hippocampal place cell activity. However, the primary purpose of this manuscript was to inform the community about an artefact that can be avoided using alternative approaches.

In summary, this report shows that common AAV hippocampal injection procedures of Ca²⁺ indicators may lead to aberrant Ca²⁺ micro-waves in wildtype mice and genetic mouse models of disease, particularly if high titre virus loads are used. The aim of this article is not to discredit Ca²⁺ indicators expressed under the synapsin promoter, a tool that we greatly appreciate ourselves, but to sensitize the field to artefactual transduction-induced aberrant Ca²⁺ micro-waves. The underlying mechanisms, some of which we have described above, are likely multi-faceted. This article seeks to inform and alert others to carefully evaluate their Ca²⁺ indicator expression approach for in vivo Ca²⁺ imaging of the hippocampus, which is becoming increasingly popular. There is certainly a much greater number of safe alternate hippocampal Ca²⁺ indicator viral

expression approaches than has been reported here, and we encourage others to report on viral Ca^{2+} indicator transduction safety profiles. Indeed, others have also encountered these artefactual events as recent social media posts attest (Application Specialist Team 2023 [↗](#)). With more indicators of brain cell activity becoming available (Ca^{2+} indicators and others including voltage indicators) as well as routes for viral delivery (Grødem *et al.* 2023 [↗](#)), the open and timely reporting of transduction safety profiles will reduce unnecessary animal experiments, save laboratory resources and time in future investigations into hippocampal function in health and disease.

Materials and Methods

Animals

All experiments followed the institutional guidelines of the Animal Care and Use committee of respective institutions (University of Bonn, DZNE and Columbia university). We used wild-type C57BL/6J mice, Thy1-GCaMP6 mice (C57BL/6J-Tg(Thy1-GCaMP6s)GP4.12Dkim/J; Jackson Lab Stock No: 025776, or C57BL/6J-Tg(Thy1-GCaMP6f)GP5.5Dkim/J; Jackson Lab Stock No: 024276 (Dana *et al.* 2014)), Vglut1-IRES2-Cre-D mice (Jackson Lab Stock No: 037512) crossed with Ai162(TIT2L-GC6s-ICL-tTA2)-D mice (Jackson Lab Stock No: 031562), PV-Cre::APPswe/PS1dE9 (cross between B6;129P2-Pvalbtm1(cre)Arbr/J, Jackson Lab Stock No: 008069, and B6.Cg-Tg(APPswe,PSEN1dE9)85Dbo/Mmjax, Jackson Lab Stock No: 034832) or *Scn2a*^{A263V} mice (from Dirk Isbrandt see Schattling *et al.* 2016 [↗](#)). Mice were kept under a light schedule of 12h on/12h off, constant temperature of $22 \pm 2^\circ\text{C}$, and humidity of 65%. They had ad libitum access to water and standard laboratory food at all time. All efforts were made to minimize animal suffering and to reduce the number of animals used.

Virus injections

For in vivo two-photon imaging experiments, GECIs were virally transduced using injection of an AAV (see **Table 1** [↗](#) and **Table 2** [↗](#)). At the time of injection mice ranged in age from 5 – 79 wks. There was no significant relationship between the age of the animal and the incidence nor frequency of Ca^{2+} micro-waves during this period (linear regression fit to the Ca^{2+} wave frequency against age was not significant: intercept = 1.37, slope = -0.007, $p=0.62$, $n = 14$; and generalized linear model relating Ca^{2+} wave incidence ~ age was not significant: z score = 0.19, deviance above null = 0.04, $p = 0.85$, $n=24$).

At the IEECR/University of Bonn: Mice (~6 wks of age) received ketoprofen (Gabrilen, Mibe; 5 mg/kg body weight; injection vol. 0.1 ml/10 g b.w., s.c.) for analgesia and anti-inflammatory treatment 30 minutes prior to induction of anesthesia. Then, mice were anesthetized with 2-3% isoflurane in an oxygen/air mixture (25/75%) and then placed in a stereotactic frame. Eyes were covered with eye-ointment (Bepanthen, Bayer) to prevent drying and body temperature was maintained at 37°C using a regulated heating plate (TCAT-2LV, Physitemp) and a rectal thermal probe. After hair removal and superficial disinfection, a drop of 10% lidocaine was used to locally anesthetise the area. After 3-5 mins, a flap of skin was removed about 1 cm^2 around the middle of the skull. Residual soft tissue was then removed from the skull with a scraper and 3% $\text{H}_2\text{O}_2/\text{NaCl}$ solution. After complete drying, cranial sutures served as landmarks for the determination of injection sites. For virus injection, a burr hole was carefully drilled through the skull using a dental drill, avoiding excessive heating and injury to the meninges by intermittent cooling with sterile PBS. Coordinates were: for CA1, anterioposterior [AP] measured from bregma 1.9 mm, lateral [L] specified from midline 1.6 mm, dorsoventral [DV] from surface of the skull 1.6 mm; for dentate gyrus, AP 2.4 mm; L 1.6 mm; x3 injections at DV 2.7, 2.5, 2.1 mm. Virus particles (see **Table 1** [↗](#)) were slowly injected (20-100 nl/min). To prevent reflux of the injected fluid upon cannula retraction, it was left in place until 5 min post-injection, and then carefully lifted.

Construct	Full plasmid	AAV	Source	ID
Syn.GCaMP6s	Syn.GCaMP6s.WPRE.SV40	AAV1	Addgene	100843
Syn.GCaMP6m	Syn.GCaMP6m.WPRE.SV40	AAV1	Addgene	100841
Syn.jGCaMP7f	Syn.jGCaMP7f-WPRE	AAV9	Addgene	104488
Syn.GCaMP6f	Syn.GCaMP6f.WPRE.SV40	AAV1	Addgene	100837
hSyn1.R-CaMP1.07	hSyn1-chI-RCaMP1.07-WPRE-SV40p(A)	AAV9	VVF/UZH	V224-9
syn.Flex.GCaMP6s	Syn.Flex.GCaMP6s.WPRE.SV40	AAV1	Addgene	100845
syn.Flex.GCaMP6m	Syn.Flex.GCaMP6m.WPRE.SV40	AAV1	Addgene	100838
CamKII0.4.Cre.SV40	CamKII 0.4.Cre.SV40	AAV1 or 9	Addgene	105558

Table 2

Key Resources

At the DZNE: A more detailed procedure was described previously (Fuhrmann *et al.* 2015 [↗](#); Poll *et al.* 2020 [↗](#)). Briefly, mice (6–78 wks) were anesthetized with i.p. injection of Ketamine (0.13 μ g/g) and Xylazine (0.01 μ g/g), head-fixed using a head holder (MA-6N, Narishige, Tokyo, Japan) and placed into a motorized stereotactic frame (Luigs-Neumann, Ratingen, Germany). Body temperature was constantly controlled by a self-regulating heating pad (Fine Science Tools, Heidelberg, Germany). After skin incision and removal of the pericranium, position of the injection 34_JG cannula was determined in relation to bregma. A 0.5 μ m hole was drilled through the skull (Ideal micro drill, World Precision Instruments, Berlin, Germany). Stereotactic coordinates were taken from Franklin and Paxinos, 2008 (The Mouse Brain in Stereotaxic Coordinates, Third Edition, Academic Press). Virus (see **Table 1** [↗](#)) was injected in two loci with the following CA1 coordinates: AP 1.95 mm; L 1.5 mm; DV 1.15 mm at a speed of 100 nL/min.

At Columbia University: A more detailed procedure was described previously (Wenzel *et al.* 2017 [↗](#), 2019a [↗](#)). Briefly, mice (8–20 wks) were anesthetized with isoflurane (initial dose 2–3% partial pressure in air, then reduction to 1–1.5%). For viral injections, a small cranial aperture was established using a dental drill above somatosensory cortex (coordinates from bregma: AP 2.5 mm, L 0.24 mm, DV 0.2 mm), or V1 (coordinates from lambda: AP =2.5 mm, L = 0.02 mm, DV 0.2–0.3 mm), or the hippocampus (coordinates from bregma, CA1: -1.9mm, -1.6mm, -1.6mm, CA3: -2.2 mm, -2.3 mm, -2.7 mm). A glass capillary pulled to a sharp micropipette was advanced with the stereotaxic instrument, and virus particles (see **Table 1** [↗](#)) were injected into putative layer 2/3 of neocortex over a 5 min period at 50 nL/min, or hippocampus over 12.5 min using a UMP3 microsyringe pump (World Precision Instruments).

At University of Zurich: A more detailed procedure was described previously (Rupprecht *et al.* 2023 [↗](#)). Briefly, mice (18 wks) were anesthetized using isoflurane (5% in O₂ for induction, 1–2% for maintenance during surgery) and provided with analgesia (Metacam 5 mg/kg bodyweight, s.c.). Body temperature was maintained at 35–37°C using a heating pad. An incision was made into the skin after local application of lidocaine. Viral particles (see **Table 1** [↗](#)) were injected into CA1 (coordinates AP -2.0 mm, ML -1.5 mm from Bregma, DV -1.3) using a glass pipette with a manually driven syringe at a rate of approximately 50 nL/min. The injection pipette was left in place for further 5 mins before being slowly retracted.

In vivo imaging window implantation procedure

In vivo imaging window implantation procedure: Cranial window surgery was performed to allow imaging from the dorsal hippocampal CA1/CA3 region or neocortex.

At the IEECR, University of Bonn: 30 minutes prior to induction of anesthesia, buprenorphine was administered for analgesia (Buprenovet, Bayer; 0.05 mg/kg body weight; injection vol. 0.1 ml/20 g b.w., i.p.). Further, dexamethasone (Dexa, Jenapharm; 0.1 mg/20 g b.w.; injection vol. 0.1 ml/20 g b.w., i.p.), and ketoprofen (Gabrilen, Mibe; 5 mg/kg b.w.; injection vol. 0.1 ml/10 g b.w., s.c.) were applied to counteract inflammation, swelling and pain. Mice were anesthetized with 2–3% isoflurane in an oxygen/air mixture (25/75%) and then placed in a stereotactic frame. Eyes were covered with eye-ointment (Bepanthen, Bayer), body temperature was maintained at 37°C by closed-loop regulation through a warming pad (TCAT-2LV, Physitemp) and a rectal thermal probe. Throughout the course of the surgical procedure, the isoflurane dose was successively reduced to about 1–1.5% at a gas flow of ~0.5 ml/minute. A circular craniotomy ($\varnothing$ ~3 mm) was established above the right hemisphere/hippocampus within the central opening ($\varnothing$ ~7 mm) of the head plate using a dental drill. Cortical tissue was carefully aspirated until the alveolar fibers above CA1 could be visually identified. A custom-made silicon cone (top $\varnothing$ 3 mm, bottom $\varnothing$ 2 mm, depth 1.5 mm, RTV 615, Momentive) attached to a cover glass ($\varnothing$ 5 mm, thickness 0.17 mm) was inserted and fixed with dental cement around the edges of the cover glass (see Masala *et al.* 2022 [↗](#)). Postoperatively, all mice received analgetic treatment by administration of buprenorphine twice daily (Buprenovet, Bayer; 0.05 mg/kg b.w.; injection vol. 0.1 ml/20 g b.w., i.p.) and ketoprofen once

daily (Gabrilen, Mibe; 5 mg/kg b.w.; injection vol. 0.1 ml/10 g b.w., s.c.) for 3 consecutive days post-surgery. Throughout this time, animals were carefully monitored twice daily. Animals typically recovered from surgery within 24-48 hours, showing normal activity and no signs of pain or distress.

At the DZNE, Prior to surgery mice were anesthetized with an intraperitoneal injection of ketamine/xylazine (0.13/0.01 Jmg_Jper_Jg of body weight). Additionally, an anti-inflammatory (dexamethasone, 0.2 Jmg_Jper_Jkg) and an analgesic drug (buprenorphine hydrochloride, 0.05 Jmg_Jper_Jkg; Temgesic, Reckitt Benckiser Healthcare) were subcutaneously administered. A cranial window (Ø 3 mm) was implanted above the right hippocampus as previously described (Poll *et al.* 2020 [DOI](#)).

At Columbia University: For neocortical imaging, directly following virus injection, the craniotomy was covered with a thin glass cover slip (3×3mm, No. 0, Warner Instruments), which was fixed in place with a slim meniscus of silicon around the edge of the glass cover and finally cemented on the skull using small amounts of dental cement around the edge. For hippocampal imaging, a small area of cortex (around 1.5×1.5mm) above left CA1 was removed by gentle suction down to the external capsule, as described previously (Dombeck *et al.* 2010 [DOI](#); Wenzel *et al.* 2019b [DOI](#)). The site was repeatedly rinsed with sterile saline until no further bleeding could be observed. Then, a small UV-sterilized miniature glass plug (1.5×1.5mm, BK7 glass, obtained from BMV Optical), glued to the center of a thin glass coverslip (3×3mm, No. 0, Warner Instruments) with UV-sensitive glue, was carefully lowered onto the external capsule, and until the edges of the attached glass cover touched the skull surrounding the craniotomy. Finally, the plug was fixed in place with a slim meniscus of silicon around the edge of the glass cover and by applying small amounts of dental cement around the edge of the glass cover.

At University of Zurich: A more detailed procedure was described previously (Rupprecht *et al.* 2023 [DOI](#)). Briefly, 2 weeks after virus injection, the hippocampal window was implanted. Two layers of light-curing adhesive (iBond Total Etch, Kulzer) were applied to the exposed skull, followed by a ring of dental cement (Charisma, Kulzer). A 3-mm diameter ring was drilled into the skull, centered at the previous injection site. The cortex in the exposed region was carefully aspirated using a vacuum pump until the stripes of the corpus callosum became visible. The corpus callosum was left intact. A cylindrical metal cannula (diameter 3 mm, height 1.2-1.3 mm) attached with dental cement to a coverslip (diameter 3 mm) was carefully inserted into the cavity. The hippocampal window was fixed in place using UV-curable dental cement (Tetric EvoFlow, Ivoclar).

Two-photon Ca^{2+} imaging

A variety of standard commercially available two-photon systems were used at the different institutes to record the Ca^{2+} micro-waves.

At the IEECR, University of Bonn: a commercially available two-photon microscope was used (A1 MP, Nikon), equipped with a 16x water-immersion objective (N.A.=0.8, WD=3 mm, CFI75 LWD 16X W, Nikon), and controlled by NIS-Elements software (Nikon). GCaMP6s was excited at 940 nm using a Ti:Sapphire laser system (~60 fs laser pulse width; Chameleon Vision-S, Coherent). Emitted photons were collected using gated GaAsP photomultipliers (H11706-40, Hamamatsu). Several individual tif series were recorded by resonant scanning at a frame rate of 15 Hz for a total 20-40 minutes per imaging session.

At the DZNE: Recordings of Ca^{2+} -changes were performed with a galvo-resonant scanner (Thorlabs, Newton, USA) on a two-photon microscope equipped with a 16x water immersion objective with a numerical aperture of 0.8 (N16XLWD-PF, Nikon, Düsseldorf, Germany) and a titanium sapphire (Ti:Sa) 80 MHz Chameleon Ultra II two-photon laser (Coherent, Dieburg,

Germany) that was tuned to 920 nm for GCaMP6m fluorescence excitation. GCaMP6m fluorescence emission was detected using a band-pass filter (525/50 nm, AHF, Tübingen, Germany) and a GaAsP PMT (Thorlabs, Newton, USA). ThorImageLS software (Thorlabs, version 2.1) was used to control image acquisition. Image series (896 x 480 pixels, 0.715 $\mu\text{m}/\text{pixel}$, or 640 x 256 pixels) were acquired at 30.3 Hz or 32.3 Hz.

At Columbia University: Neural population activity was recorded using a commercially available two-photon microscope (Bruker; Billerica, MA) and a Ti:Sapphire laser (Chameleon Ultra II; Coherent) at 940 nm through a 25x objective (Olympus, water immersion, N.A. 1.05). Resonant galvanometer scanning and image acquisition (frame rate 30.206 fps, 512×512 pixels) were controlled by Prairie View Imaging software.

At University of Zurich: Neuronal population activity was recorded using a custom-built two-photon microscope (see [Rupprecht et al. 2023](#) [↗](#)). Briefly light from a femtosecond-pulsed laser (MaiTai, Spectra physics; tuned to 960 nm; power below objective 40-45 mW) was used to scan the sample below a 16x objective (Nikon, water immersion, NA 0.8). Image acquisition and scanning (frame rate 30.88Hz, 622×512 pixels) were controlled by custom-written software (Chen et al. 2016).

Analysis of aberrant Ca^{2+} micro-waves

To remove motion artifacts, recorded movies were registered using a Lucas–Kanade model (Greenberg & Kerr 2009 [↗](#)), or the ImageJ plugin moco (available through the Yuste web page, or <https://github.com/NTCColumbia/moco> [↗](#)) (Dubbs et al. 2016 [↗](#)) or in the case of R-CaMP the NoRMCorre algorithm (Pnevmatikakis & Giovannucci 2017 [↗](#)).

We determined the diameter of the calcium waves in a semi-automated fashion from the raw tif-series. Using ImageJ software, we first drew an orthogonal line across the largest aspect of each calcium wave progressing through the field of view, which resulted in a fluorescent histogram for each wave. Using custom code (MATLAB R2020b), we further analysed all histograms for each mouse and imaging timepoint. First, we applied a gentle smoothing, max-normalized each histogram, and max-aligned all histograms of a given imaging session. Then, after excluding the 10% lowest fluorescent values, the width of each calcium wave and a mean value was calculated for each time point/imaging session. Finally, the resulting pixel values were converted to micrometer, based on the respective objective (@ 512×512pxl and 1x Zoom: Nikon 16x, NA 0.8, 3mm WD: 1.579 $\mu\text{m}/\text{pxl}$; Olympus 25x, NA 1.05 2mm WD: 0.92 $\mu\text{m}/\text{pxl}$). The speed of the Ca^{2+} micro-waves was calculated from the duration and path length of the events visually identified and manually tracked in the FOV.

Histochemistry

To verify successful viral transduction and window position, animals were deeply anesthetized with ketamine (80 mg/kg body weight) and xylazine (15 mg/kg body weight) at the end of the respective experiment. After confirming a sufficient depth of anesthesia, mice were heart-perfused with cold phosphate buffered saline (PBS) followed by 4% paraformaldehyde (PFA) in PBS. Animals were decapitated and the brain removed and stored in 4% PFA in PBS solution for 24 hours. Fifty to 100 μm thick coronal slices of the hippocampus were cut on a vibratome (Leica). For nuclear staining, brain slices were kept for 10 min in a 1:1000 DAPI solution at room temperature. Brain slices were mounted and the green and blue channel successively imaged under an epi-fluorescence or spinning disc microscope (Visitron VisiScope).

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Author Contributions

M.W. and T.K. conceived and planned the study. N.M., M.W. M.M, E.A., D.A.O, F.D., P.R performed experiments. F.H., R.Y., M.F., H.B. provided infrastructure and experimental expertise. M.W. and T.K. analysed data and wrote the manuscript with input from all authors.

Conflict of Interest

None of the authors have a conflict of interest.

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Editors

Reviewing Editor

Jun Ding

Stanford University, Stanford, United States of America

Senior Editor

Laura Colgin

University of Texas at Austin, Austin, United States of America

Reviewer #2 (Public Review):

Summary:

The authors describe and quantify a phenomenon in the CA1 and CA3 of the hippocampus that they call aberrant Ca²⁺ micro-waves. Micro-waves are sometimes seen during 2-photon calcium imaging of populations of neurons under certain conditions. They are spatially confined slow calcium events that start in a few cells and slowly spread to neighboring groups of cells. This phenomenon has been uttered between researchers in the field at conferences, but no one has taken the time to carefully capture and quantify micro-waves and pin down the causes. The authors show that micro-waves are dependent on the viral titre of the genetically encoded calcium indicators (GECIs), the genetic promoter (synapsin), the neuronal subtype (granule cells in the dentate gyrus do not produce micro-waves and they are not seen in the neocortex), and the density of GECI expression. The authors should be commended for their work and for raising awareness to all labs doing any form of calcium imaging in populations of neurons. The authors also come up with alternative approaches to avoid artifactual micro-waves such as reducing the transduction titre (1:2 dilution of virus) and a transduction method employing sparser and cre-dependent GECI expression in principal cells using a CaMKII promoter.

Strengths:

The micro-waves reported in the paper were robustly observed across 4 laboratories and 3 different countries with various experimenters and calcium imaging set-ups. This adds significant strength to the work.

The age of mice used covered a broad range (from 6 to 43 weeks). This is a strength because it covers most ages that are used in labs that regularly do calcium imaging.

Another strength is they used different GCaMP variants (GCaMP6m, GCaMP6s, GCaMP7f), as well as a red indicator: RCaMP. This shows the micro-waves are not an issue with any particular GECI, as the authors suggest.

The authors include many movies of micro-waves. This is extremely useful for researchers in the field to view them in real-time so they can identify them in their own data.

They provide a useful table with specific details of the virus injected, titre, dilution, and other information along with the incidence of micro-waves. A nice look-up table for researchers to see if their viral strategy is associated with a high or low incidence of micro-waves.

Weaknesses:

The effect of micro-waves on single cell function was not analyzed. It would be useful, for example, if we knew the influence of micro-waves on place fields. Can a place cell still express a place field in a hippocampus that produces micro-waves? What effect might a microwave passing over a cell have on its place field? Mice were not trained in these experiments, so the authors do not have the data. However, they do briefly discuss these ideas.

<https://doi.org/10.7554/eLife.93804.2.sa2>

Reviewer #3 (Public Review):

Summary:

The work by Masala and colleagues highlights a striking artifact that can result from a particular viral method for expressing genetically encoded calcium indicators (GECIs) in neurons. In a cross-institutional collaboration, the authors find that viral transduction of GECIs in the hippocampus can result in aberrant slow-traveling calcium (Ca^{2+}) micro-waves. These Ca^{2+} micro-waves are distinct from previously described ictal activity but nevertheless are likely a pathological consequence of overexpression of virally transduced proteins. Ca^{2+} micro-waves will most-likely obscure the physiology that most researchers are interested in studying with GECIs, and their presence indicates that the neural circuit is in an unintended pathological state. Interestingly this pathology was not observed using the same viral transduction methods in other brain regions. The authors recommend several approaches that may help other experimenters avoid this confound in their own data such as reducing the titer of viral injections or using recombinase-dependent expression. The intent of this manuscript is to raise awareness of the potential unintended consequences of viral overexpression, particularly for GECIs. A rigorous investigation into the exact causes of Ca^{2+} micro-waves or the mechanisms supporting them are beyond the authors' intended scope.

Strengths:

The authors clearly demonstrate that Ca^{2+} micro-waves occur in the CA1 and CA3 regions of the hippocampus following large volume, high titer injections of adeno-associated viruses (AAV1 and AAV9) encoding GECIs. The supplementary videos provide undeniable proof of their existence.

By forming an inter-institutional collaboration, the authors demonstrate that this phenomenon is robust to changes in surgical techniques or imaging conditions.

Weaknesses:

I believe that the weaknesses of the manuscript are appropriately highlighted by the authors themselves in the discussion. The manuscript does not attempt to exhaustively characterize the conditions under which calcium micro-waves occur. Rather, the authors raise awareness of this problem.

<https://doi.org/10.7554/eLife.93804.2.sa1>

Reviewer #4 (Public Review):

Summary:

Masala N et al showed interesting aberrant calcium microwaves in the hippocampus when synapsin promoter driven GCaMPs were expressed for a long period of time. These aberrant hippocampal Ca²⁺ micro-waves depend on the viral titre of the GECI. The microwave of Ca²⁺ was not observed when GECI was expressed only a sparse set of neurons.

Strengths:

These findings are important to wide neuroscience community especially when considering a great number of investigators are using similar approaches. Results look convincing and are consistent across several laboratories.

Weaknesses:

Synapsin promoter labels both excitatory pyramidal neurons and inhibitory neurons. To avoid aberrant Ca²⁺ microwave, a combination of Flex virus and CaMKII-Cre or Thy-1-GCaMP6s and 6f mice were tested. However, all these approaches limit the number of infected pyramidal neurons. While the comprehensive display of these results is appreciated, one additional important test would be more informative. To distinguish whether the microwave of Ca²⁺ is sufficiently caused via the expression of GCaMP in interneurons, or just a matter of pyramidal neuron density, testing Flex-GCaMP6 in interneuron specific mouse lines such as PV-Cre and SOM-Cre will provide further clarifications.

<https://doi.org/10.7554/eLife.93804.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Weaknesses:

One important question is needed to further clarify the mechanisms of aberrant Ca²⁺ microwaves as described below.

Synapsin promoter labels both excitatory pyramidal neurons and inhibitory neurons. To avoid aberrant Ca²⁺ microwave, a combination of Flex virus and CaMKII-Cre or Thy-1-GCaMP6s and 6f mice were tested. However, all these approaches limit the number of infected pyramidal neurons. While the comprehensive display of these results is appreciated, a crucial question remains unanswered. To distinguish whether the microwave of Ca²⁺ is caused selectively via the abnormality of interneurons, or just a

matter of pyramidal neuron density, testing Flex-GCaMP6 in interneuron specific mouse lines such as PV-Cre and SOM-Cre will be critical.

We agree that unravelling the role of interneurons is important to the understanding of the cellular mechanisms. However, the primary goal of this preprint was to alert the field and those embarking on in vivo Ca²⁺ imaging to AAV transduction induced artefacts mediated by one of the most widely used viral constructs for Ca²⁺ imaging in the field. It was important to us to distribute this finding among the community in a timely manner to avoid the unnecessary waste of resources.

We consider a thorough understanding of cell-type specific mechanisms interesting. However, the biological relevance of the Ca²⁺ waves is as yet unclear and to disentangle exactly which cellular and subcellular factors that drive the aberrant phenomenon will require a large systematic effort which goes beyond our resources. For instance, it will be technically not trivial to separate biologically relevant contributions from technical differences. For instance, the absence of Ca²⁺ waves under the principal neuron promotor CaMKII may suggest the involvement of interneurons. However, alternate possibilities are a reduced density of expression across principal neurons or that the expression levels between the 2 promoters is different.

The important, take-home message of the preprint, in our opinion, is that users check carefully their viral protocols, adjust the protocols for their specific scientific question and report any issues. We now emphasise the fact that although Ca²⁺ waves were not observed following conditional expression of syn.GCaMP with CaMKII.cre, this may not be due to a requirement for interneuronal expression but simply reflect differences in final GCaMP expression density and levels between the two transduction procedures (P12, L298-303).

Reviewer #2 (Public Review):

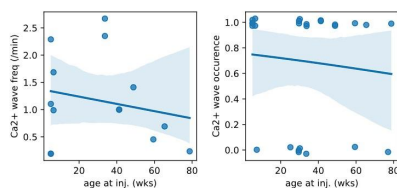
Weaknesses:

Whether micro-waves are associated with the age of mice was not quantified. This would be good to know and the authors do have this data.

We plotted the animal age at the time of injection for all injections of Syn.GCaMP6 into CA1/CA3 and found no correlation in either the occurrence of Ca²⁺ waves nor the frequency of Ca²⁺ waves during the age period between 5 – 79 wks (see reviewer Fig1; linear regression fit to the Ca²⁺ wave frequency against age was not significant: intercept = 1.37, slope = -0.007, p=0.62, n = 14; and generalized linear model relating Ca²⁺ wave ~ age was not significant: z score = 0.19, deviance above null = 0.04, p = 0.85, n=24). We have now added a statement to this in the revised manuscript (P14 L354-359) and for the reviewers we have added the plots below.

Author response image 1.

Plot of Ca²⁺ micro-wave frequency (left: number of Ca²⁺ waves/min) or occurrence (right: yes/no) against the animal age at the time of viral injection. Blue line is linear (left) or logistic (right) fit to the data with 95% confidence level.



The effect of micro-waves on single cell function was not analyzed. It would be useful, for example, if we knew the influence of micro-waves on place fields. Can a place cell still express a place field in a hippocampus that produces micro-waves? What effect might a microwave passing over a cell have on its place field? Mice were not trained in these experiments, so the authors do not have the data.

We agree that these are interesting questions; however, the preprint is focused on describing the GECI expression conditions prone to generating these artefacts. Studying the effects of Ca²⁺ micro-waves on the circuitry are scientific questions, and would require an experimental framework of testing the aberrant activity on a specific physiological function e.g. place activity or specific oscillations (e.g. sharp-wave activity). Ca²⁺ microwaves, as the ones described here, have not been reported under physiological conditions or pathophysiological conditions and studying the effects of such artefactual waves on the circuit was not our intention.

With respect to place cell activity, specifically, it is intuitive that during the Ca²⁺ micro-wave the participating cell's place field activity would be obscured by the artefactual activity. Cell activity appears to return immediately following the wave suggesting that the cells could exhibit place activity outside their participation in the Ca²⁺ micro-waves. However, we do not know if the Ca²⁺ micro-wave activity disrupts the generation or maintenance of place fields. We have now added a brief reference to possible effects on place coding to the paper (P12, L315-317).

The CaMKII-Cre approach for flexed-syn-GCaMP expression shows no micro-waves and is convincing, but it is only from 2 animals, even though both had no micro-waves. In light of the reviewer's comment, we have added a further 3 animals with conditional expression of GCaMP6m from the DZNE to complement the current dataset with conditional expression of GCaMP6s from UoB (P10, L236 & 239 and revised table 1). Although Ca²⁺ waves were not observed in any of the in total 5 animals, we still do not know with all certainty whether this approach is completely safe. Time will show if researchers still encounter the phenotype under certain conditions when using this conditional approach.

The authors state in their Discussion that even without observable microwaves, a syn-Ca²⁺-indicator transduction strategy could still be problematic. This may be true, but they do not check this in their analysis, so it remains unknown

We agree with the reviewer and have now made this point clearer in the revised discussion (P11, L257-258)

Reviewer #3 (Public Review):

Weaknesses:

I believe that the weaknesses of the manuscript are appropriately highlighted by the authors themselves in the discussion. I would, however, like to emphasize several additional points.

As the authors state, the exact conditions that lead to Ca²⁺ micro-waves are unclear from this manuscript. It is also unclear if Ca²⁺ micro-waves are specific to GECI expression or if high-titer viral transduction of other proteins such as genetically encoded voltage indicators, static fluorescent proteins, recombinases, etc could also cause Ca²⁺ micro-waves.

The high expression of other proteins has been shown to result in artefactual phenomenon such as toxicity or fluorescent puncta (for GFP see Hechler et al. 2006; Katayama et al. 2008 for GEVI see Rühl et al. 2021), but we are not aware of reports of micro-waves. Although it is certainly possible that high expression levels of other proteins could lead to waves, we suspect the Ca²⁺ micro-waves observed in this preprint result from a dysregulation of Ca²⁺ homeostasis. This is not to suggest that voltage indicators could not result in micro-waves (e.g. Ca²⁺ homeostasis may be indirectly affected).

The authors almost exclusively tested high titer ($>5 \times 10^{12}$ vg/mL) large volume (500-1000 nL) injections using the synapsin promoter and AAV1 serotypes. It is possible that Ca²⁺ micro-waves are dramatically less frequent when titers are lowered further but still kept high enough to be useful for in vivo imaging (e.g. 1×10^{12} vg/mL) or smaller injection volumes are used. It is also possible that Ca²⁺ micro-waves occur with high titer injections using other viral promoter sequences such as EF1a or CaMKIIa. There may additionally be effects of viral serotype on micro-wave occurrence.

We agree with all points raised by the reviewer. Notably, we used viral transduction protocols with titers and volumes within in the range of those previously used for viral transduction of GCaMP under the synapsin promoter (see P11 L269-275) and we observed Ca²⁺ micro-waves. As the reviewer suggested, we did find that lowering the titer is an important factor in reducing these Ca²⁺ micro-waves and there is likely a wide range of approaches that avoid the phenomenon. With regards to viral serotype, we show that micro-waves occurred across AAV1 and 9, but it is possible that other serotypes may avoid the phenomenon.

We reiterate in the abstract of the revised manuscript that expression level is a crucial factor (P2, L40 and P2, L44-45) and now mention that other promoters and induction protocols that result in high Ca²⁺ indicator expression may result in Ca²⁺ micro-waves (P12, L291-294).

The number of animals in any particular condition are fairly low (Table 1) with the exception of V1 imaging and thy1-GCaMP6 imaging. This prohibits rigorous comparison of the frequency of pathological calcium activity across conditions.

We have now added 3 more animals with conditional GCaMP6 expression. In total, the study contains 34 animals with viral injection into the hippocampus from different laboratories and under different conditions resulting in multiple groups. As such we are cognizant of the resulting limitations for statistical evaluation.

However, in light of the reviewer's comment, we have now employed a generalized linear model tested on all the data to examine the relationship between the Ca²⁺ micro-wave incidence and the different factors. The multivariate GLM did find a significant relationship between Ca²⁺ micro-wave incidence and both viral dilution and weeks post injection (see below and revised manuscript P8, L189-193).

For injections into CA1 in the hippocampus (n=28), a GLM found no relationship between Ca²⁺ micro-waves and each of the individual variables x (Ca-wave ~ x) ; viral dilution: z score = 1.14, deviance above null = 1.31, p = 0.254; post injection weeks: : z score = 1.18, deviance above null = 1.44, p = 0.239; injection volume: : z score = -0.76, deviance above null = 0.59, p = 0.45; construct: : z score = 1.18, difference in deviance above null = 1.44, p = 0.239)

However, a multivariable logistic GLM relating dilution and post injection weeks (Ca-wave ~ dilution + p.i_wks) showed that together both variables were significantly related to Ca²⁺ micro-waves (Deviance above null = 7.5; Dilution: z score = 2.18, p < 0.05; p.i_wks : z score = 2.22, p < 0.05).

Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

Results are straightforward and convincing. While a couple of ways to reduce the aberrant microwaves of calcium responses were demonstrated, delving into the functions of interneurons is crucial for a more comprehensive understanding of cellular causality.

As mentioned in the public response, disentangling cellular mechanism from technical requirements will need a large and systematic study. To determine the contribution from interneurons, the use of specific interneuron promoters would be required, and viral titers systematically varied to result in similar cellular GCaMP expression levels as seen under the synapsin promoter condition.

Reviewer #2 (Recommendations For The Authors):

Do the authors think the cells are firing when they participate in a micro-wave, or do they think the calcium influx is due to something else? A discussion point on this would be good.

This is an excellent point raised by the reviewer. We do not know if the elevated cellular Ca²⁺ during the artifactual Ca²⁺ micro-wave reflects action potential firing or an increase of Ca²⁺ from intracellular stores. As already described in the text of the preprint, their optical spatiotemporal profile neither fits with known microseizure progression patterns, nor with spreading depolarization/depression. We have adopted the reviewer's suggestion and added the following point to the discussion section in the revised preprint (P12, L308-315):

In a limited dataset, we attempted to detect the Ca²⁺ micro-waves by hippocampal LFP recordings (using a conventional insulated Tungsten wire, diameter ~110µm). We could not identify a specific signature, e.g. ictal activity or LFP depression, which may correspond to these Ca²⁺ micro-waves. The crucial shortcoming of this experiment of course is that with these LFP recordings, we could not simultaneously perform hippocampal 2-photon microscopy. Thus, it is uncertain if the Ca²⁺ micro-waves indeed occurred in proximity to our electrode.

The results seem to suggest that micro-waves may involve interneurons as their CaMKII-Cre strategy avoids waves - possibly due to a lack of expression of GECIs in interneurons. It would be great to hear the author's thoughts on this and add a brief discussion point.

As mentioned in public response to Reviewer 1, it is difficult to disentangle cellular mechanisms from technical requirements, and the exact requirements for the Ca²⁺ micro-waves to occur are still not fully clear. The absence of Ca²⁺ micro-waves in our CaMKII-Cre dataset may indeed reflect the requirement of interneurons. However, it could just as well be due to a sparse labelling of principle cells or simply reflect differences in the expression levels of GCaMP under the different promoters.

All in all, a more complete understanding of the requirements of such Ca²⁺ micro-waves will require a community effort. Therefore, it is important that each group check the safety profile of their GECI and report problems to the community.

We have added these points to the revised preprint (P12, L291 and P12, L298)

Plotting the incidence of micro-waves as a function of the age of mice would be a nice addition (the authors have the data).

There was no relationship of Ca²⁺ micro-wave occurrence or frequency with age over the range of 5-79 wks (see public response) and this has been added to the preprint (P14, L354)

Reviewer #3 (Recommendations For The Authors):

I appreciate the authors raising the awareness of this issue. I had personally observed micro-waves in my own data as well. In agreement with their findings, I found that the occurrence of micro-waves was dramatically lower when I reduced the viral titer. Anecdotally, I also observed voltage micro-waves when virally transducing genetically encoded voltage indicators at similar titers. For that reason, I am skeptical that this issue is exclusive to GECIs.

We find it interesting that the reviewer has also seen artefactual micro-waves following viral transduction of genetically encoded voltage indicators. Without seeing the voltage waves the referee is referring to or the conditions, it is of course difficult to compare with the Ca²⁺ micro-waves we report. However, this comment again raises the question of mechanism. We believe that in the GECI framework, Ca²⁺ homeostatic aspects are important. Voltage indicators are based on different sensor mechanisms, and expressed in the cell membrane, but it may very well be that there are overlapping factors between Ca²⁺ and voltage indicators that could trigger a similar, or even the same phenomenon in the end.

Minor comments:

(1) Line 131-132: I believe the authors only tested for micro-waves in V1. This should be made clear in the results. It could be that micro-waves could occur in other parts of cortex with the same viral titers.

Both V1 and somatosensory cortex were tested as described in the methods (P15, L395-397), we have made this clearer in the revised preprint (P6, L138).

(2) There are no statistics associated with the data from Fig 1e.

We have now added statistics (P5, L126).

(3) The authors may be able to make a stronger claim about the pathological nature of the micro-waves if there are differences in the histology between the injected and non-injected hemispheres. For example, is there evidence of widespread cell death in the injected hemisphere (e.g. lower cell count, smaller hippocampal volume, caspase staining, etc).

We found no evidence of gross morphological changes to the hippocampus following viral transduction with no changes in CA1 pyramidal cell layer thickness or CA1 thickness (pyramidal cell layer thickness: 49 ± 12.5 μ m ipsilateral and 50.3 ± 11.1 μ m contralateral, n=4, Student's t-test p=0.89; CA1 thickness: 553.3 ± 14 μ m ipsilateral and 555.8 ± 62 μ m contralateral, n = 4, Student's t-test p=0.94; 48 $\pm$ 13 weeks post injection at time of perfusion).

We have added this to the preprint (P5, L117-122)

(4) The broader micro-waves in the stratum oriens versus the stratum pyramidale are likely due to the spread of the basal dendrites of pyramidal cells. If the typical size of the basal dendritic arbor of CA1 pyramidal neurons is taken into account, does this explain the wider calcium waves in this layer.

Absolutely, great point, yes, we completely agree on this. It is likely the active neuropil (including dendritic arbour) are contributing to the apparent broader diameter. In addition, as evident in the video 5 cell somata in the stratum Oriens (possibly interneurons) are active and their processes also contribute.

We have now mentioned these points in the revised preprint (P5, L132)

| (5) Lines 179-181: *Is the difference in the prevalence of micro-waves between viral titers statistically significant?*

Although we have a large number of animals in total (n=34) with viral injection into the hippocampus, the number of animals in each condition, given the many factors, is low. We therefore used a generalized linear model to test the relationship between the Ca²⁺ micro-waves and the variables.

We have now added this analysis to the revised preprint (P8, L189-193)

| (6) Lines 200-203: *The CA3 micro-waves were only observed at one institution. The current wording is slightly misleading.*

We agree and have changed this to be clearer (P9 L216)